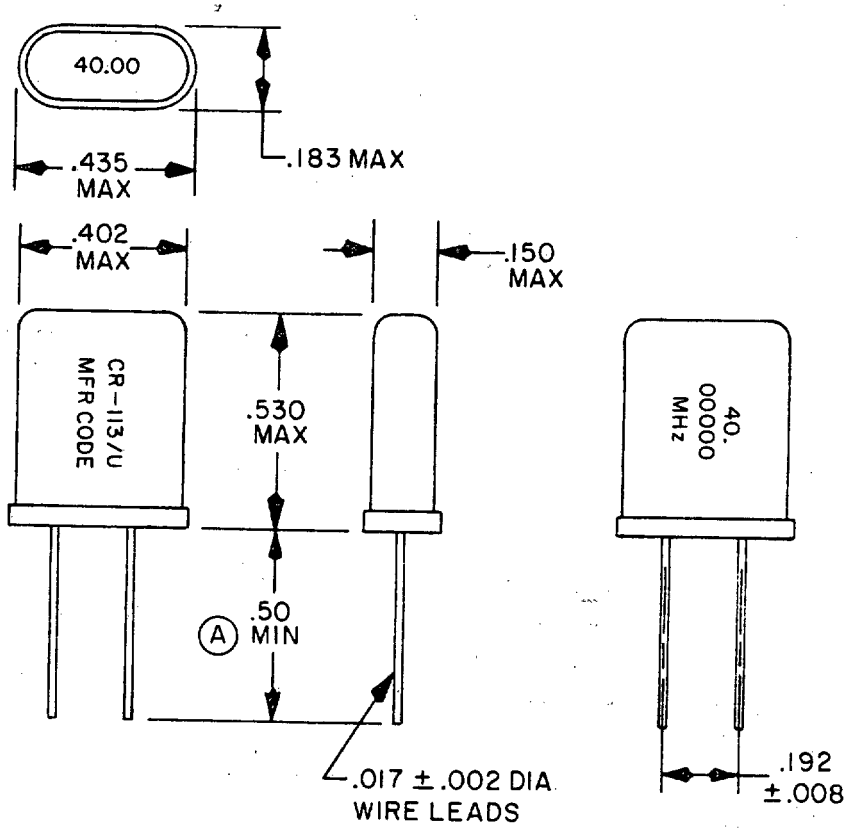
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-113/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 10 to 61 MHz; third mechanical overtone; controlled; series resonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-113/U.

Ⓐ denotes changes

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Ⓐ TABLE I. Detail requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions	See figure 1.
Frequency range	10 to 61 MHz, inclusive
Frequency tolerances:	
Operating temperature range	±10 ppm
Room temperature	±70 ppm
Frequency stability	±2.5 ppm
Mode of oscillation	Third mechanical overtone
Reference temperature	75°C +1°C
Temperature ranges:	
Operating (controlled)	70°C to 80°C, inclusive
Operable	80°C to 90°C, inclusive
Government reference standard test set	TS-683/TSM
Resonance	Series
Calibration values and rated drive level	See table II
Capacitance, shunt	7 pF, maximum
Equivalent resistance	40 ohms, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10
Aging:	
Frequency change permitted	±5 ppm

1/ Method 201 of MIL-STD-202.

TABLE II. Calibration values and rated drive level.

Band of specified frequencies, inclusive	Calibration values		Rated drive level
	Resistance	Resistor voltage drop	
MHz	Ohms	Volt	mW
10 to 15	60	0.35	2.0 ± 0.4
15+ to 25	40	0.28	2.0 ± 0.4
25+ to 61	40	0.20	1.0 ± 0.2

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 85
DLA - ES

User activities:
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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